

Chapter 4: Convolutional Neural Networks (CNNs)

A Comprehensive Guide to Spatial Feature Extraction and Computer Vision

1. Motivation for CNNs in Computer Vision

Standard Fully Connected (Dense) networks struggle with image data because flattening a 2D image (e.g., 256x256x3) creates an enormous number of parameters (nearly 200,000 weights per hidden neuron), leading to severe overfitting and computational bottlenecks. Furthermore, dense networks discard spatial 2D relationships between neighboring pixels. CNNs solve these challenges through three foundational principles: local receptive fields, shared weights (parameter sharing), and spatial subsampling.

2. The Convolution Operation and Filters

The core building block of a CNN is the convolutional layer. A set of learnable kernels (filters), typically 3x3 or 5x5, slides across the input tensor. At each location, element-wise multiplication is performed followed by a summation, producing a 2D activation map (feature map). Key parameters include stride (step size of filter movement) and padding (adding zeros around borders to maintain spatial dimensions, known as 'same' padding).

3. Pooling Layers (Downsampling)

Pooling layers reduce the spatial dimensions (height and width) of feature maps while retaining essential information. Max Pooling takes the maximum value in a window (e.g. 2x2 with stride 2), providing translation invariance and drastically reducing parameter count and computational complexity. Average Pooling calculates the mean value across the window.

4. Landmark Architectures: LeNet to ResNet

Over the years, landmark CNN architectures have driven progress in computer vision: LeNet-5 (1998, digit recognition), AlexNet (2012, ImageNet breakthrough using ReLU and Dropout), VGGNet (2014, deep stacks of small 3x3 convolutions), and ResNet (2015, residual skip connections solving the degradation and vanishing gradient problem in ultra-deep networks of 50-152 layers).

5. Advantages of CNNs

The primary advantages of CNNs mentioned in this chapter are: (a) Parameter sharing reduces model complexity compared to dense layers; (b) Translation invariance allows detecting objects regardless of position; (c) Hierarchical feature learning automatically extracts edges in early layers, textures in middle layers, and complex semantic objects in deep layers without manual feature engineering.